HERIOT WATT UNIVERSITY

Department of Mathematics

Mock Exam – Solutions
F18CE2 - Multivariable Calculus and Real Analysis B
Semester 2, 2021/2022

Attempt all questions
Show any calculations you make
(no marks will be given if there is no, or incorrect, working, even if the answer is correct).

1. Are all bounded sequences convergent? Provide a proof or a counterexample.

Solution: No, for instance $\{(-1)^n\}$ and $\{\sin(n)\}$ are bounded sequences that are not convergent.

[2]

State the two conditions that define the infimum.

Solution: α is the infimum of a set A if α is a lower bound and if it is the largest lower bound.

[2]

Let A be a set of positive real numbers that is bounded above, and consider the set

$$B = \left\{-7a^2 + 1 : a \in A\right\}.$$

Show that B is bounded below and that $\inf B = -7(\sup A)^2 + 1$.

Solution: Let $\alpha = \sup A$, then define $\beta = -7\alpha^2 + 1$. We need to show that $\beta = \inf B$. For any $b = -7a^2 + 1 \in B$ satisfies $b = -7a^2 + 1 \geq -7\alpha^2 + 1 = \beta$, so β is a lower bound. To show that it is the largest lower bound, consider $\{a_n\}$ a sequence in A such that $\lim_{n\to\infty} a_n = \alpha$ (which exists by Theorem 1.43), then show that $\lim_{n\to\infty} -7a_n^2 + 1 = -7\alpha^2 + 1 = \beta$.

[4]

2. Consider the sequence $\{a_n\}$ given by

$$a_n = \frac{7^n}{(2n+1)!}.$$

Prove that $\{a_n\}$ is decreasing.

Solution: By the ratio test,

$$\frac{a_{n+1}}{a_n} = \frac{7}{(2n+3)(2n+2)} = \frac{7}{4n^2 + 10n + 6} < \frac{7}{4n + 10n} = \frac{1}{2} < 1,$$

the sequence is decreasing.

[4]

Use this to show that a_n converges, stating fully any theorem you use.

Solution: Monotone Convergence Theorem (MCT): any monotone bounded sequence is convergent.
The sequence $\{a_n\}$ is bounded below by 0, therefore is convergent by the MCT.

[4]

3. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous at $a \in \mathbb{R}$ and $\lim_{x\to a} f(x) = L > 0$. Prove that there exists a $\delta > 0$ such that $f(x) > 0$ for any $x \in (a - \delta, a + \delta)$.

Solution: In the definition of continuity at $x = a$, take $\varepsilon = L/2$. Then, by definition, there exists a $\delta = \delta(L/2)$ such that for any $x \in \mathbb{R}$ with $|x - a| < \delta$ we have

$$|f(x) - L| < L/2.$$

Hence, for any $x \in (a - \delta, a + \delta)$, $f(x) > L - L/2 = L/2 > 0$.

[5]

4. Use the Intermediate Value Theorem to prove that the equation

$$e^{-x} = x + x^2$$

has a real solution in $[0, 1]$.

Solution: Define $f(x) = e^{-x} - x - x^2$ on $\mathbb{R}$. Then f is continuous and differentiable on $\mathbb{R}$. Also,

$$f(0) = 1, \quad f(1) = e^{-1} - 2.$$

Since $f(1) < 0 < f(0)$, by the IVT, there exists a point $a \in (0, 1)$ such that $f(a) = 0$.

[4]

Then use Rolle's theorem to show that it cannot have more than one solution on $\mathbb{R}$.

Solution: If there were two solutions, say $a, b \in \mathbb{R}$, with $a < b$, then $f(a) = f(b) = 0$. Since the function f is differentiable on $[a, b]$, we can apply Rolle's theorem. Therefore, there exists a point $c \in [a, b]$ such that $f'(c) = 0$. In order to have a contradiction, we need to prove that f' is never 0 on $\mathbb{R}$. One way to do so, is to observe that $f'(-\ln 2) < 0$ and f' is increasing for $x < -\ln 2$ and decreasing for $x > -\ln 2$, therefore it is never 0 (in particular, this is an application of the MVT).

[5]

5. Use the First Mean Value Theorem to prove that for any integer $n \geq 1$,

$$(n+1)e^n \leq (n+1)e^{n+1} - ne^n \leq (n+2)e^{n+1}.$$

Solution: Mean Value Theorem (MVT): Given a function f that is continuous on $[a,b]$ and differentiable on (a,b), there exists a point $c \in (a,b)$ such that $f(b) - f(a) = f'(c)(b-a)$.
Define the function xe^x on $\mathbb{R}$. This is continuous and differentiable on $\mathbb{R}$. For a fixed integer $n \geq 1$, we apply the MVT to f on $[n, n+1]$, therefore there exists a $c \in (n, n+1)$ such that

$$(n+1)e^{n+1} - ne^n = (1+c)e^c.$$

Since $n < c < n+1$, we have that $(n+1)e^n < (1+c)e^c < (n+2)e^{n+1}$.

[7]

Note: State clearly any theorems you are using, where you are using the result of these theorems and why they are applicable.

6. Let $\{a_n\}$ be a sequence of real numbers such that $Ar^{-1/2} \leq a_n \leq Br^{-1/2}$ for some constants $A, B > 0$. Discuss the convergence of

(a) $\displaystyle\sum_{r=1}^{\infty} a_r$,

(b) $\displaystyle\sum_{r=1}^{\infty} a_r^2$,

(c) $\displaystyle\sum_{r=1}^{\infty} \sqrt{r} a_r^2$,

Solution: Comparison Test: Given two sequences $\{a_n\}, \{b_n\}$ such that $0 \leq a_n \leq b_n$ for any $n \geq 1$, we have that

- if $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges too,
- if $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges too.

(a) We have that $a_n \geq Ar^{-1/2} \geq 0$, and $\sum_{n=1}^{\infty} r^{-1/2}$ diverges. Therefore $\sum_{n=1}^{\infty} a_n$ diverges too.

(b) We have that $a_n^2 \geq A^2 r^{-1} \geq 0$, and $\sum_{n=1}^{\infty} r^{-1}$ diverges. Therefore $\sum_{n=1}^{\infty} a_n^2$ diverges too.

(c) We have that $\sqrt{n} a_n^2 \geq A^2 r^{1/2} r^{-1} = A^2 r^{-1/2}$. Similarly as in (i), the series diverges.

[6]

stating clearly any theorem you are using.

7. Find the radius of convergence R of the power series

$$\sum_{r=1}^{\infty} \frac{2^r\, x^r}{r^2 + \sqrt{r}}\,.$$

Solution: Let $p_r = \frac{2^r}{r^2+\sqrt{r}}$. By the ratio test,

$$\lim_{r\to\infty} \frac{p_{r+1}}{p_r} = \lim_{r\to\infty} \frac{2(r^2+\sqrt{r})}{(r+1)^2+\sqrt{r+1}} = 2,$$

the radius of convergence is $R = 1/2$, by Corollary 4.22.

[4]

8. Use the definition of the Riemann integral $\displaystyle\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}\,dx$ to prove that

$$\lim_{n\to\infty} \left(\frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \frac{1}{\sqrt{4n^2-9}} + \cdots + \frac{1}{\sqrt{4n^2-n^2}} \right) = \frac{\pi}{6}.$$

Solution: Consider the equi-partition of $[0, 1/2]$ given by $P_n = \left\{0, \frac{1}{2n}, \frac{2}{2n}, \frac{3}{2n}, \ldots, \frac{1}{2}\right\}$ and the continuous function $f(x) = \frac{1}{\sqrt{1-x^2}}$ on $[0, 1/2]$. Then,

$$\begin{aligned} U(P_n, f) &= \sum_{r=0}^{n-1} \delta_n f\left(\frac{r}{2n}\right) \\ &= \sum_{r=1}^{n} \frac{1}{2n} \frac{1}{\sqrt{1 - r^2/4n^2}} \\ &= \sum_{r=1}^{n} \frac{1}{2n} \frac{2n}{\sqrt{4n^2 - r^2}} \\ &= \sum_{r=1}^{n} \frac{1}{\sqrt{4n^2 - r^2}}. \end{aligned}$$

Since the function f is continuous (and monotone) on $[0, 1/2]$, and $\delta_n \to 0$ as $n \to \infty$, by Corollary 5.18, $\lim\limits_{n\to\infty} U(P_n, f) = \displaystyle\int_0^{1/2} f = \pi/6$.

[8]

Note: You can assume that $\displaystyle\int_0^{1/2} \frac{1}{\sqrt{1-x^2}}\,dx = \frac{\pi}{6}$ *without computing it.*

9. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} x^n, & x < 0, \\ x^{n+1}, & x \geq 0. \end{cases}$$

Find the smallest positive integer n for which f is twice differentiable on $\mathbb{R}$.

Solution: For $n = 1$, we have that

$$f(x) = \begin{cases} x, & x < 0, \\ x^2, & x \geq 0, \end{cases}$$

therefore

$$f'(x) = \begin{cases} 1, & x < 0, \\ 2x, & x \geq 0. \end{cases}$$

Taking the side limits, we see that f' is not continuous at $x = 0$, therefore it cannot be differentiable. For $n = 2$, we have that

$$f(x) = \begin{cases} x^2, & x < 0, \\ x^3, & x \geq 0, \end{cases} \qquad f'(x) = \begin{cases} 2x, & x < 0, \\ 3x^2, & x \geq 0, \end{cases} \qquad f''(x) = \begin{cases} 2, & x < 0, \\ 6x, & x \geq 0. \end{cases}$$

Hence, f'' exists and f is differentiable. For a general $n \geq 2$, we have that

$$f(x) = \begin{cases} x^n, & x < 0, \\ x^{n+1}, & x \geq 0, \end{cases} \qquad f'(x) = \begin{cases} nx^{n-1}, & x < 0, \\ (n+1)x^n, & x \geq 0, \end{cases}$$

$$f''(x) = \begin{cases} n(n-1)x^{n-2}, & x < 0, \\ (n+1)nx^{n-1}, & x \geq 0, \end{cases}$$

therefore f is twice differentiable, as f'' exists.

Observe that f'' is not continuous for $n = 2$. If we wanted to have f'' continuous, we would need $n \geq 3$.

[7]

Note: You can use the usual rules of calculus to compute the first and second derivatives of the function, without proving their validity.

10. Let (a_n) be a sequence defined by

$$a_n := \left(\frac{1}{2n}\right)^n \qquad \text{for } n \geq 1\,.$$

(a) Find $c > 0$ and $0 < q < 1$ such that

$$a_n \leq cq^{n-1} \qquad \text{for all } n \geq 1\,.$$

Solution: $a_n = \left(\frac{1}{2n}\right)^n = \left(\frac{1}{2n}\right)\left(\frac{1}{2n}\right)^{n-1} \leq \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^{n-1}$, for any $n \geq 1$, hence a possible choice is $c = q = 1/2$.

(b) Show with (a) and the geometric progression $\sum_{n=1}^{\infty} cq^{n-1} = \frac{c}{1-q}$ that there exists a constant $U > 0$ such that

$$b_N = \sum_{n=1}^{N} a_n \leq U\,.$$

Solution: The series $\displaystyle\sum_{n=1}^{\infty} cq^{n-1}$ converges, which means that the sequence of the partial sums $S_N = \displaystyle\sum_{n=1}^{N} cq^{n-1}$ is convergent, therefore S_N is bounded, and there exists a U such that $S_N \leq U$. Since $a_n \leq cq^{n-1}$, we have that $b_N \leq S_N \leq U$.

(c) Proof with the Monotone Convergence Theorem based on (a) and (b) that the sequence of partial sums (b_N) converges for $N \to \infty$, that means that the series

$$\sum_{n=1}^{\infty} a_n$$

converges.

Solution: We proved that $\{b_N\}$ is bounded above, and now we prove that it is increasing. By the MCT, this proves that it converges (i.e. the series $\displaystyle\sum_{n=1}^{\infty} a_n$ converges).
In order to show that $\{b_N\}$ is increasing, we observe that

$$b_{N+1} - b_N = a_{N+1} = \left(\frac{1}{2(N+1)}\right)^{N+1} > 0.$$

[8]

End of paper.